Notes E

Tuesday, October 6, 2026 10:48 PM

Integral of $f \geq 0$. (mble) (not necessarily simple)

To define $\int f \, d\mu$, the steps are — | $E(x) = ?$ $x \geq 0$.

(A) - get hold of a seq. of nonneg. simple fns. $\{f_n\} \uparrow f$. Calculate $\int f_n \, d\mu \quad \forall\, n \geq 1$

(B) - define $\int f \, d\mu := \lim_{n \to \infty} \int f_n \, d\mu$. (:= defn)

Qn. (1) How do we know such $\{f_n\}$ exists? ||?

yes! and can be (multiple) such.

(see discussion below rem. 2.3.1)

Qn. (2) what if there are more than one such seq? $\{f_n\}$ and $\{g_n\} \uparrow f$. || ??

(well-defined?)

(it's ok - P. 2.3.2)

then we'll verify basic properties. (Cor. 2.3.3)

+ other properties : MCT, Fatou's lemma.

proof of p.2.3.2 : $D_n = \{\omega : f_n(\omega) \geq \rho g_N(\omega)\}$

$$\int f_n \, d\mu = \int \left(f_n 1_{D_n} + f_n 1_{D_n^c} \right) d\mu \quad \checkmark$$

$f_n 1_{D_n}$, $f_n 1_{D_n^c}$: simple fns. ≥ 0.

$$= \int f_n 1_{D_n} \, d\mu + \int f_n 1_{D_n^c} \, d\mu$$

by linearity of int. of simple fns.

by linearity of int. of simple fns.

$\int f_n 1_{D_n^c} \, d\mu \geq 0$. mono. of int. of simple fns.

$$\geq \int f_n 1_{D_n} \, d\mu \quad \checkmark$$

on D_n : $f_n > \rho g_N$, $D_n = \{f_n > \rho g_N\}$

$$f_n^{(\omega)} 1_{D_n}^{(\omega)} \geq \rho g_N^{(\omega)} 1_{D_N}^{(\omega)}$$

$$\int f_n 1_{D_n} \, d\mu \geq \int \rho g_N 1_{D_n} \, d\mu \quad \text{(by mono.)}$$

$$= \rho \int g_N 1_{D_n} \, d\mu \quad \text{(lin.)}$$

$\omega \in D_n$ ✓

$\Rightarrow \omega \in D_{n+1}$ ✓

or $D_n \uparrow \bigcup_{n \geq 1}^{\infty} D_n$

$D_n =$

$= \{f_n - \rho g_N > 0\}$

$= (f_n - \rho g_N)^{-1}(0, \infty)$

$\in \mathcal{F}$ ✓

In prop 2.3.2: $f_n, g_n \geq 0$, simple $\uparrow f$ (a.e.(μ))

then $\lim_{n\to\infty} \int f_n \, d\mu = \lim_{n\to\infty} \int g_n \, d\mu$.

In MCT (Thm. 2.3.4): $f_n \uparrow f$, a.e. (μ), $f \geq 0$.

then $\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu = \lim_{n\to\infty} \int g_n \, d\mu$.

where g_n simple ≥ 0, $g_n \uparrow f$.

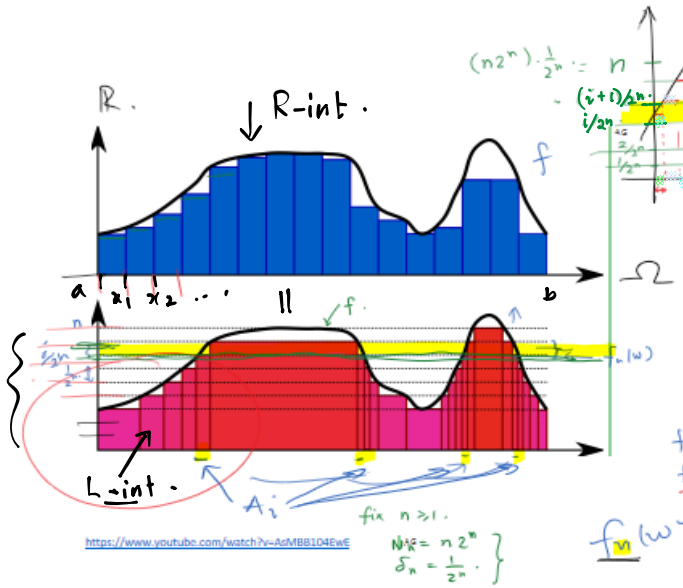
But here $f_n \geq 0$ (not necc. simple)

proof: $f_n \uparrow f$ a.e. (μ).

i.e $\exists$ A s.t $\forall \omega \in A$, $f_n(\omega) \uparrow f(\omega)$, and $\mu(A^c) = 0$

$\tilde{D}_n = \{ f_n \geq \rho g_k \} \to D_n = \tilde{D}_n \cap A \uparrow (\Omega \cap A) = A$.

(as before).



https://www.youtube.com/watch?v=AsMBB1O4EwE

$(n2^n)\cdot\frac{1}{2^n} = n$

$(i+1)/2^n$

$i/2^n$

$f_n^{(\omega)} = n$ if $f(\omega) \geq n$.

$= \frac{i}{2^n}$, if $\frac{i}{2^n} \leq f(\omega) < \frac{i+1}{2^n}$

if has at most $n2^n$ values. (simple f_n)

$f_2 \geq f_1$

$f_{n+1} \geq f_n$

$f_n \uparrow f$

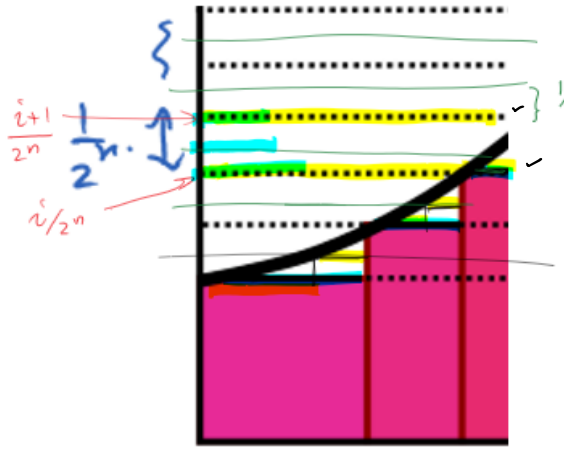
fix $n \geq 1$.

$N_n = n2^n$

$\delta_n = \frac{1}{2^n}$.

$f_n(\omega) = \frac{i}{2^n}$, if $\omega \in A_i$

$A_i = \{\omega : \frac{i}{2^n} \leq f(\omega) < \frac{i+1}{2^n}\}$



$1/2^{n+1} = \frac{1}{2}\cdot\frac{1}{2^n}$.

$f_{n+1}(\omega) \geq f_n(\omega)$ ✓

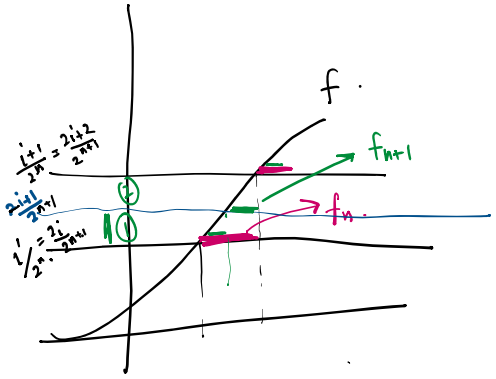
if $\omega \in \left[\frac{i}{2^n}, \frac{i+1}{2^n}\right]$.

$\left[\frac{2i}{2^{n+1}}, \frac{2(i+1)}{2^{n+1}}\right]$ ✓

$\left[\frac{2i}{2^{n+1}}, \frac{2i+1}{2^{n+1}}\right] \cup \left[\frac{2i+1}{2^{n+1}}, \frac{2i+2}{2^{n+1}}\right]$

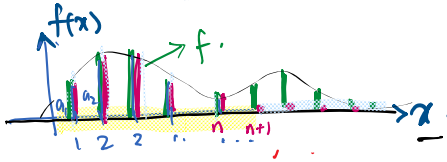
① ②

$\mathcal{C} = \{[a,b] \ldots\}$



$(\Omega, \mathcal{F}, \mu) = (\mathbb{N}, \mathcal{P}(\mathbb{N}), \mu = \text{ctng meas})$

$f(x) = a_x$, $x \in \mathbb{N}$. $\quad a_x = \frac{1}{x^2}$.



$f_n = f\,1_{(1,n]}$.

$f_n(x) = \begin{cases} f(x) & \text{if } x \le n \\ 0 & \text{if } x > n \end{cases}$

$f_n \uparrow f$.

$f_n(x) = \sum_{i \le x} a_i \, 1_{\{i\}}(x)$ simple.

$\int f_n \, d\mu \quad \sum_{i \le n} a_i \, \mu(\{i\}) = \sum_{i=1}^{n} a_i$

$\lim_{n\to\infty} \int f_n \, d\mu = \sum_{i=1}^{\infty} a_i \, \mu(\{i\}) = \sum_{i=1}^{\infty} a_i$

$\int f \, d\mu$ $\quad \lim_{n\to\infty} \sum_{i=1}^{n} a_i \, \mu(\{a_i\}) = \sum_{i=1}^{\infty} a_i$

$\int f \, d\mu = \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$

Cor. 2.3.3. (idea):

Suppose $f, g \geq 0$, $\alpha, \beta \geq 0$.

Show $\int (\alpha f + \beta g)\, d\mu = \alpha \int f\, d\mu + \beta \int g\, d\mu$. } linearity

So for f, $\exists \{f_n\} \uparrow f$ (simple), $\exists \{g_n\} \uparrow g$ (simple).

Then $h_n = \{\alpha f_n + \beta g_n\}$ is a seq. of simple fns $\uparrow (\alpha f + \beta g)$

$$\int (\alpha f + \beta g)\, d\mu = \lim_{n\to\infty} \int (\alpha f_n + \beta g_n)\, d\mu \quad (h_n \text{ simple})$$

$$= \lim_{n\to\infty} \left[\alpha \int f_n\, d\mu + \beta \int g_n\, d\mu \right]$$

(by lin. prop. of int. of simple fns)

$h_n \uparrow (\alpha f + \beta g)$

$$= \alpha \int f\, d\mu + \beta \int g\, d\mu.$$

— x —

Fact: (Recall: $\int f \cdot 1_A \, d\mu = 0$ if $\mu(A) = 0$ & $f \geq 0$ simple fn. (checked))

→ if $f \geq 0$, $\exists A \in \mathcal{F}$ s.t. $\mu(A) = 0$ Then

$$\int f \, d\mu = \int f 1_{A^c} \, d\mu, \quad \int f 1_A \, d\mu = 0.$$

$$\int f \, d\mu = \int f 1_A \, d\mu + \int f 1_{A^c} \, d\mu.$$

$X \geq 0$ a.s.(P), $E(X 1_A) = 0$ if $P(A) = 0$. ✓

Claim: $f \cdot 1_A = 0$ a.e.(μ). $\Rightarrow \int f 1_A \, d\mu = 0$

proof: $\{\omega : f(\omega) 1_A(\omega) \neq 0\} \subseteq \{\omega : 1_A(\omega) \neq 0\} = A$.

$\mu(\,\downarrow\,) \leq \mu(A) = 0$

$\Rightarrow f 1_A = 0$ a.e.(μ).

$\Rightarrow \int f 1_A \, d\mu = \int 0 \cdot d\mu = 0.$

$\Rightarrow \int f 1_A \, d\mu = 0$ $(= \int 0 \cdot d\mu)$

$\int f \, d\mu = \int f 1_{A^c} \, d\mu$ where $f \geq 0$, $\mu(A) = 0$.

$a \cdot b \neq 0 \Rightarrow a \neq 0$ & $b \neq 0$

$a \cdot b = 0 \Leftarrow a = 0$ OR $b = 0$

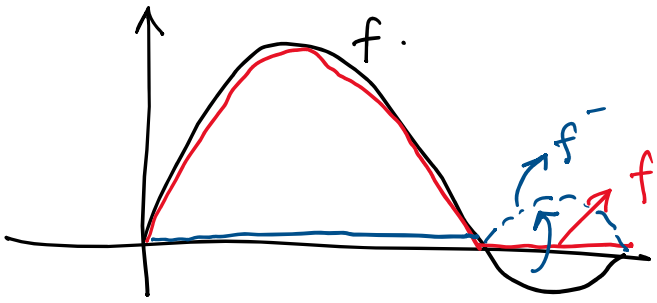
* Int. of gen. (meas.) f. on $(\Omega, \mathcal{F}, \mu)$.

$\forall\, x \in \mathbb{R}$; $x^+ = x\, I_{(x \geq 0)}$, $x^- = -x \cdot I_{(x<0)}$

$x = x^+ - x^-$ | $x^+ + x^- = |x|$.

For any general f, define $f^+ = f\, 1_{\{f \geq 0\}} \geq 0$

$f^- = -f\, 1_{\{f<0\}} \geq 0$.



$\int f^+ d\mu$, $\int f^- d\mu$ as earlier

Since, $f = f^+ - f^-$,

we define $\int f d\mu = \int f^+ d\mu - \int f^- d\mu$, (to preserve linearity)

if at least $\int f^+ d\mu < \infty$ or $\int f^- d\mu < \infty$.

But we say f is "integrable" if both are finite.

in that case

$\Leftrightarrow \quad \int |f| \, d\mu = \int f^+ \, d\mu + \int f^- \, d\mu$. (since $|f(\omega)| = f^+(\omega) + f^-(\omega)$)

$< \infty$ $\quad < \infty$ $\quad < \infty$.

$\therefore \quad < \infty$.

* $\int f \, d\mu = \pm\infty$ possible, but if $\int |f| \, d\mu < \infty$ then $\int f \, d\mu < \infty$ and we say f is "integrable"

Recall: if $(\Omega, \mathcal{F}, P)$ then X is r.v.
(prob.)

$\int X \, dP$.

$E(X) = E(X^+) - E(X^-) = \pm\infty$ possible.

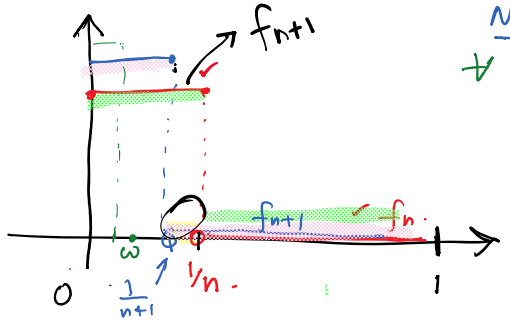
X is $(\Omega, \mathcal{F}, P)$-integrable if $E(|X|) < \infty$.

In this case we also say $f \in L_1(\Omega, \mathcal{F}, \mu)$

in gen. $f \in L_p$ if $\int |f|^p d\mu < \infty$.

$E(f_n) =$

Rem. 2.3.6: $f_n(\omega) = n\, I_{[0,\frac{1}{n}]}(\omega)$, $n \geq 1$. $([0,1], \mathcal{B}[0,1], m)$



Note: $f_n \to f \equiv 0$ a.e.(m).

$\forall\, \omega \in (0,1]$, $f_n(\omega) = 0$ for large n. $= f(\omega)$

$f_n(0) = n \quad \forall n \geq 1. \to \infty$. $(\neq f(\omega))$

$m(\{\omega : f_n(\omega) \not\to f(\omega)\}) = m(\{0\}) = 0$

So $f_n \to f$ a.e.(m).

$f_n(0) = n \uparrow \infty$.

$f_n \to f \quad \forall\, \omega \neq 0$.

$\boxed{f_n \to f \text{ a.e.}(m)}$ ($f = 0$)

$E(f_n) = \int f_n \, dm$

$= n \cdot m((0, \frac{1}{n}])$

$= n \cdot \frac{1}{n} = 1$.

MCT: $\int f_n \, d\mu = n \cdot m([0, \frac{1}{n}]) = n \cdot \frac{1}{n} = 1$.

$f_n \not\to f$

$\int f \, d\mu = \int 0 \, d\mu = 0$.

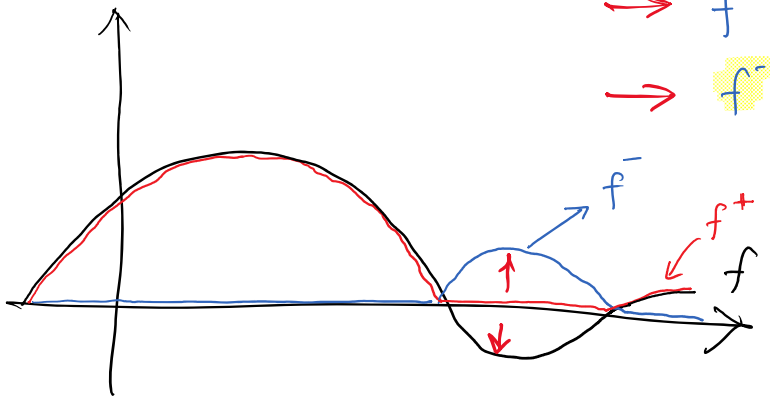
$\lim_{n\to\infty} \int f_n \, d\mu \neq \int f \, d\mu$.

$\lim \int f_n \, d\mu > \int (\lim f_n - f_n) \, d\mu$

example ">"

Fatou: $\liminf \int f_n \, d\mu = 1 \geq \int \liminf(f_n) \, d\mu = 0$. ($\liminf(f_n) = 0$ a.e.)

The inequality can be strict.



$f^+ = f\, I_{(f \geq 0)}$ ✓ ≥ 0

$f^- = (-f)\, I_{(f < 0)}$ ≥ 0.

$f = f^+ + f^-$

$|f| = f^+ + f^-$

note.

we know how to compute $\left\{ \int f^+ d\mu, \int f^- d\mu. \right.$

but either can be $+\infty$.

Define $\int f d\mu = \int f^+ d\mu - \int f^- d\mu$. ✓

as long as both are NOT $+\infty$ (don't want $\infty - \infty$)

But we say f is integrable if

$\int f d\mu < \infty$. or $\int f^+ d\mu < \infty$ AND $\int f^- d\mu < \infty$.

Or $\int |f| d\mu = \int f^+ d\mu + \int f^- d\mu < \infty$.

$|f| = f^+ + f^-$
$\geq 0 \quad \geq 0 \quad \geq 0$.

So f is called integrable if $\int |f| d\mu < \infty$. ←

Space of all μ-integrable fns is called

$f : \Omega \to \mathbb{R}$
$\mathcal{F}, \mu$.

$L^1(\Omega, \mathcal{F}, \mu) \equiv L^1(\mu) = \{f : \int |f| d\mu < \infty\}$ ✓

(in terms of r.v's $L_1(P) = \{X : E|X| < \infty\}$.)

$L^p = \{f : \int |f|^p d\mu < \infty\}$

Fact: f mble f_n, Then $\left|\int f\,d\mu\right| \le \int |f|\,d\mu$. $\left(|E(X)| \le E(|X|)\right)$

Pf: $\left|\int f\,d\mu\right| = \left|\int f^+\,d\mu - \int f^-\,d\mu\right|$

$|E(X)| \le E|X|$

$\le \left|\int f^+\,d\mu\right| + \left|\int f^-\,d\mu\right|$ ($\int f^+\,d\mu \ge 0$, $\int f^-\,d\mu \ge 0$)

$|a-b| \le |a|+|b|$

$= \int f^+\,d\mu + \int f^-\,d\mu$.

$= \int (f^+ + f^-)\,d\mu = \int |f|\,d\mu$.

In particular,

$\left|\int (f-g)\,d\mu\right| \le \int |f-g|\,d\mu$.

(h.)

In DCT, $\int f_n\,d\mu \to \int f\,d\mu$?

$\left|\int f_n\,d\mu - \int f\,d\mu\right| = \left|\int (f_n - f)\,d\mu\right|$

$\le \int |f_n - f|\,d\mu \to 0$.

if $f_n \to f$ in $L_1(\Omega, \mathcal{F}, \mu)$ i.e $\int |f_n - f| d\mu \to 0$.

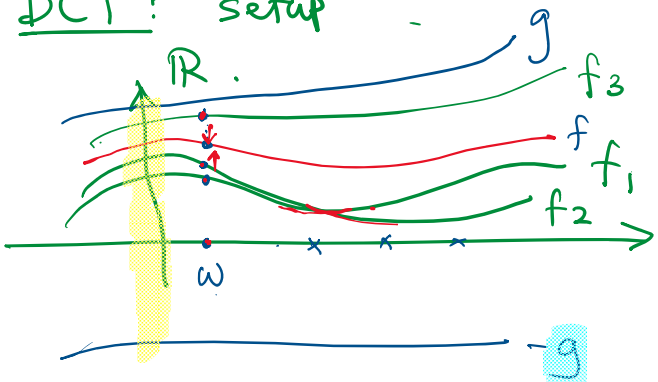
$\Rightarrow$

$$\int f_n d\mu \longrightarrow \int f d\mu \quad \text{as } n \to \infty.$$

(i.e $E(|X_n - X|) \to 0 \Rightarrow E(X_n) \to E(X)$) $\Leftarrow$

L_1-conv. conv. in mean.

DCT: setup



f_n mble.

$f_n \to f$ a.e.(μ).

$\mu(\{\omega : f_n(\omega) \not\to f(\omega)\}) = 0$.

when can we say

$\int f_n \, d\mu \to \int f \, d\mu$?

or $\int |f_n - f| \, d\mu \to 0$.

"$f_n \to f$ in L_1 metric"

$|f_n| \le g$. $(-g \le f_n \le g)$

$g \in L_1$ then $f \in L_1$ and holds.

DCT

For r.v $X_n \to X$ a.s.(P), and $|X_n| \le Y$ with $E(|Y|) < \infty$ then $EX_n \to EX$ and $E|X_n - X| \to 0$. (also $X \in L_1$).

Reverse Fatou's Lemma:

If $f_n \ge 0$ and $f_n \le f$, $f \in L_1$

then $\limsup_{n\to\infty} \int f_n \, d\mu \le \int (\limsup f_n) \, d\mu$.

Pf: similar to proof of DCT:

start with $\gamma_n = f - f_n \ge 0$. and apply Fatou's (check) usual.

